

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

06

10 answers · Rationale-rich · Teach from doc

Q1**A**

A. The estimated height, in cm, of the fluid at the start of the experiment

Choice A is correct. It's given that the function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the fluid in the beaker x days after the start of the experiment. For a function defined by an equation of the form $f(x) = b + mx$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, 39 represents the value of $f(x)$ when $x = 0$. Since $x = 0$ represents the start of the experiment, then the best interpretation of 39 in this context is that the estimated height, in cm, of the fluid was 39 at the start of the experiment.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. The estimated change in the height, in cm, of the fluid each day is 0.18, not 39.

Choice D is incorrect and may result from conceptual errors.

Q2

C

C.

x	gx
-1	-7
0	4
1	15

Choice C is correct. Each of the tables shows the same three values of x : -1, 0, and 1.

Substituting -1 for x in the given function yields $g(-1) = 11(-1) + 4$, or $g(-1) = -7$. Therefore, when $x = -1$, the corresponding value of $g(x)$ is -7. Substituting 0 for x in the given function yields $g(0) = 11(0) + 4$, or $g(0) = 4$. Therefore, when $x = 0$, the corresponding value of $g(x)$ is 4. Substituting 1 for x in the given function yields $g(1) = 11(1) + 4$, or $g(1) = 15$.

Therefore, when $x = 1$, the corresponding value of $g(x)$ is 15. The table in choice C shows -7, 4, and 15 as the corresponding value of $g(x)$ for x -values of -1, 0, and 1, respectively. Therefore, the table in choice C shows three values of x and their corresponding values of $g(x)$.

Choice A is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x + 11$.

Choice B is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x$.

Choice D is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 11x$.

Q3

C

C. 11

Choice C is correct. Substituting 4 for x in $g(x) = 5x + a$ gives $g(4) = 5(4) + a$. Since $g(4) = 31$, the equation $g(4) = 5(4) + a$ simplifies to $31 = 20 + a$. It follows that $a = 11$.

Choices A, B, and D are incorrect and may result from arithmetic errors.

Q4**D**

D. 84

Choice D is correct. The value of $f(x)$ when $x = -8$ can be found by substituting -8 for x in the given function. This yields $f(-8) = -3(-8) + 60$, or $f(-8) = 84$. Therefore, when $x = -8$, the value of $f(x)$ is 84.

Choice A is incorrect. This is the value of $-3 + -8 + 60$, not $-3(-8) + 60$.

Choice B is incorrect. This is the value of $-8 + 60$, not $-3(-8) + 60$.

Choice C is incorrect. This is the value of $-3 + 60$, not $-3(-8) + 60$.

Q5**D**

D. $mx = 5.7x$

Choice D is correct. It's given that Juan rides his bike at an average rate of 5.7 minutes per mile. The number of minutes it will take Juan to ride x miles can be determined by multiplying his average rate by the number of miles, x , which yields $5.7x$. Therefore, the function $mx = 5.7x$ models the number of minutes it will take Juan to ride x miles.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q6**50**

The correct answer is 50. Substituting 8 for x in the given equation yields $y = 58 + 10$, or $y = 50$. Therefore, the value of y is 50 when $x = 8$.

Q7

D

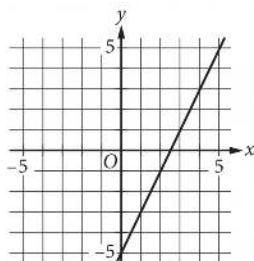
D. 88

Choice D is correct. The value of gx when $x = 8$ can be found by substituting 8 for x in the given equation $gx = 10x + 8$. This yields $g8 = 108 + 8$, or $g8 = 88$. Therefore, when $x = 8$, the value of gx is 88.

Choice A is incorrect. This is the value of x when $gx = 8$, rather than the value of gx when $x = 8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**D****D.**

Choice D is correct. In the xy -plane, the graph of the equation $y = mx + b$, where m and b are constants, is a line with slope m and y -intercept $(0, b)$. Therefore, the graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept $(0, -5)$. Having a slope of 2 means that for each increase in x by 1, the value of y increases by 2. Only the graph in choice D has a slope of 2 and crosses the y -axis at $(0, -5)$. Therefore, the graph shown in choice D must be the correct answer.

Choices A, B, and C are incorrect. The graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept at $(0, -5)$. The graph in choice A crosses the y -axis at the point $(0, 2.5)$, not $(0, -5)$, and it has a slope of $\frac{1}{2}$, not 2. The graph in choice B crosses the y -axis at $(0, -5)$;

however, the slope of this line is -2 , not 2. The graph in choice C has a slope of 2; however, the graph crosses the y -axis at $(0, 5)$, not $(0, -5)$.

Q9**A**

A. $f(x) = 3x + 29$

Choice A is correct. An equation that defines a linear function f can be written in the form $fx = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $fx = 29$. Substituting 0 for x and 29 for fx in the equation $fx = mx + b$ yields $29 = m0 + b$, or $29 = b$. Substituting 29 for b in the equation $fx = mx + b$ yields $fx = mx + 29$. It's also given in the table that when $x = 1$, $fx = 32$. Substituting 1 for x and 32 for fx in the equation $fx = mx + 29$ yields $32 = m1 + 29$, or $32 = m + 29$. Subtracting 29 from both sides of this equation yields $3 = m$. Substituting 3 for m in the equation $fx = mx + 29$ yields $fx = 3x + 29$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**B**

B. 0, -2

Choice B is correct. The y-intercept of the graph of a function in the xy -plane is the point on the graph where $x = 0$. It's given that $fx = \frac{1}{10}x - 2$. Substituting 0 for x in this equation yields $f0 = \frac{1}{10}0 - 2$, or $f0 = -2$. Since it's given that $y = fx$, it follows that $y = -2$ when $x = 0$. Therefore, the y-intercept of the graph of $y = fx$ in the xy -plane is 0, -2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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